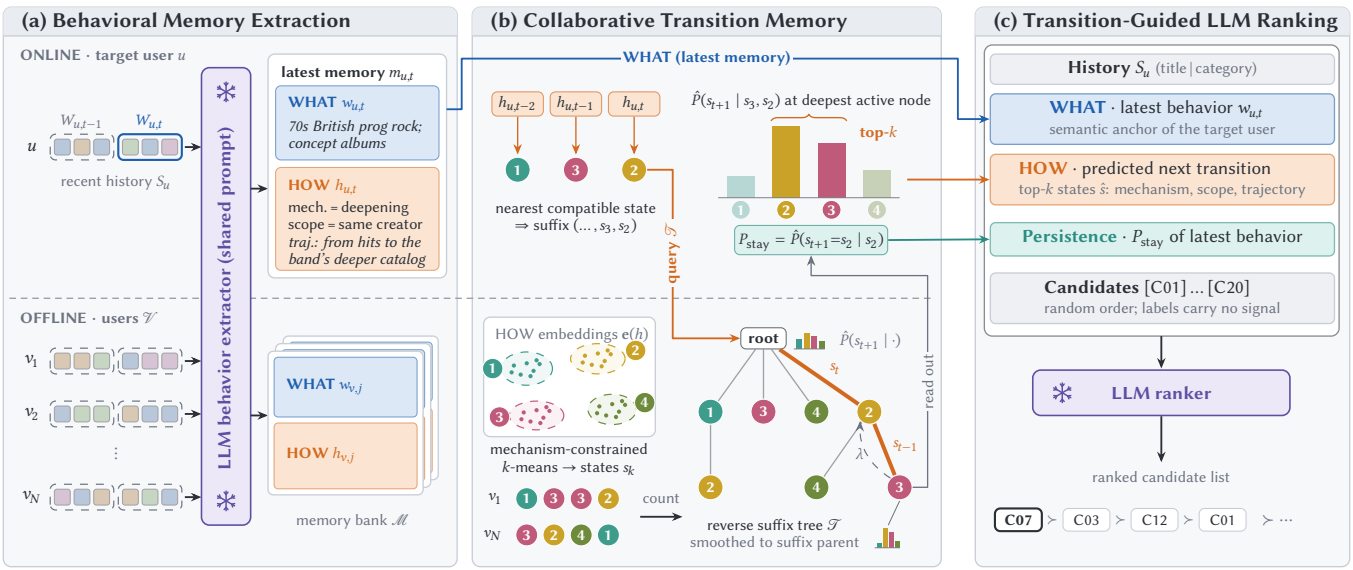
(a) Behavioral Memory Extraction
ONLINE · target user $u$
$u$
$W_{u,t-1}$
$W_{u,t}$
recent history $S_u$
LLM behavior extractor (shared prompt)
latest memory $m_{u,t}$
WHAT $w_{u,t}$
70s British prog rock; concept albums
HOW $h_{u,t}$
mech. = deepening
scope = same creator
traj.: from hits to the band's deeper catalog
OFFLINE · users $\mathcal{V}$
$v_1$
$v_2$
$v_N$
WHAT $w_{v,j}$
HOW $h_{v,j}$
memory bank $\mathcal{M}$
(b) Collaborative Transition Memory
WHAT (latest memory)
$h_{u,t-2}$
$h_{u,t-1}$
$h_{u,t}$
nearest compatible state
⇒ suffix $(\ldots, s_3, s_2)$
query $\mathcal{T}$
$\hat{P}(s_{t+1} \mid s_3, s_2)$ at deepest active node
top-$k$
$P_{\text{stay}} = \hat{P}(s_{t+1}{=}s_2 \mid s_2)$
HOW embeddings $\mathbf{e}(h)$
mechanism-constrained
$k$-means → states $s_k$
$v_1$ 1 3 3 2
$v_N$ 3 2 4 1
count
reverse suffix tree $\mathcal{T}$
smoothed to suffix parent
root
$\hat{P}(s_{t+1} \mid \cdot)$
$s_t$
$s_{t-1}$
$\lambda$
read out
(c) Transition-Guided LLM Ranking
History $S_u$ (title | category)
WHAT · latest behavior $w_{u,t}$
semantic anchor of the target user
HOW · predicted next transition
top-$k$ states $\hat{s}$: mechanism, scope, trajectory
Persistence · $P_{\text{stay}}$ of latest behavior
Candidates [C01] ... [C20]
random order; labels carry no signal
LLM ranker
ranked candidate list
C07 ≻ C03 ≻ C12 ≻ C01 ≻ ···